AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

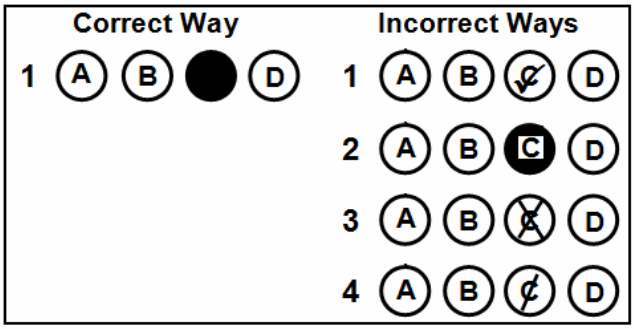
MAY 2014

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. $\frac{7}{8}$ is equal to

   A. 67.5 %
   B. 77.5 %
   C. 87.5 %
   D. 97.5 %

2. If two friends invested in a ratio of 2:3 and earned a profit of Rs. 25,000 on their investments, then the smaller share of the profit is

   A. Rs. 2,000
   B. Rs. 3,000
   C. Rs. 5,000
   D. Rs. 10,000

3. If $\frac{x}{36} = \frac{10}{9}$, then x is equal to

   A. 25
   B. 30
   C. 36
   D. 40

4. The annual saving of Mr. Saad is Rs. 200,000.Which of the following is the amount of zakat due on him, the rate of zakat being 2.5% per annum?

   A. Rs. 500
   B. Rs. 2500
   C. Rs. 5000
   D. Rs. 5500

5. What will be the amount of ushr due on a cotton yield of Rs. 265,000?

   Note: Ushr is 10% of the yield.

   A. Rs. 26.50
   B. Rs. 265.00
   C. Rs. 2,650
   D. Rs. 26,500



6. If one dozen oranges costs Rs. 100 and are sold at a price of Rs. 10 per orange, then the profit earned is

   A. Rs. 5
   B. Rs. 10
   C. Rs. 15
   D. Rs. 20

7. If a person suffers a loss of Rs. *a* and selling price is Rs. *b*, then the cost price is

A. $a + b$
B. $a - b$
C. $b - a$
D. $\frac{a+b}{2}$

8. The profit % is equal to

A. $\frac{profit}{S.P} \times 100$
B. $\frac{profit}{C.P} \times 100$
C. $\frac{profit\ \%}{S.P} \times 100$
D. $\frac{profit\ \%}{C.P} \times 100$

9. If one Saudi Riyal is approximately equal to Rs. 28, then 70 Saudi Riyals are approximately equal to

A. Rs. 1,930
B. Rs. 1,950
C. Rs. 1,960
D. Rs. 1,970

10. The time required to earn a simple interest of Rs. 1000 at an interest rate of 5% per annum on an investment of Rs. 10,000 is

A. 2 years.
B. 2.5 years.
C. 4 years.
D. 4.5 years.

11. A motor bike costs Rs. 40,000 and down payment of Rs. 10,000 is made. If the remaining amount is to be paid in 10 equal installments, then each installment is equal to

A. Rs. 3,000
B. Rs. 3,100
C. Rs. 3,200
D. Rs. 3,500

12. A company sells motorcycles on leasing. The cost of a motorcycle is Rs. 45,000 and according to the leasing plan the motorcycle is delivered on 30% down payment and 10 equal monthly installments. Each installment is of

A. Rs. 1,350
B. Rs. 3,150
C. Rs. 3,500
D. Rs. 4,500

13. If 15% sales tax is levied on a product costing Rs.200, then the consumer price is

A. Rs. 200
B. Rs. 215
C. Rs. 230
D. Rs. 245

14. A worker is paid Rs. 80 per hour for eight hours a day. After 8 hours he gets Rs. 120 for each additional hour. If he works six days for 10 hours a day, then his earning due to overtime will be

A. Rs. 1,200
B. Rs. 1,240
C. Rs. 1,400
D. Rs. 1,440

15. A tax levied by a state or province on the retail price of an item, collected by the retailer is called

A. sales tax.
B. income tax.
C. excise duty.
D. property tax.

16. A cellular company pays Rs. 200 per hour plus commission of Rs. 30 on sale of each SIM to its salesperson. If the salesperson sold 4 mobile SIMs in 3 hours, then the company will pay him

A. Rs. 320
B. Rs. 630
C. Rs. 720
D. Rs. 890

17. $\sqrt[3]{a^2}$ is equal to

A. $a^{\frac{2}{3}}$
B. $a^{\frac{3}{2}}$
C. $a^{\frac{-2}{3}}$
D. $a^{\frac{-3}{2}}$

18. $a^{-\frac{1}{2}} \times a^{\frac{3}{2}}$ is equal to

A. 1
B. $a^{\frac{1}{2}}$
C. a
D. a^2

19. The characteristic of log 0.0123 is

A. 2
B. 1
C. – 1
D. – 2

20. If the general term of a sequence is $\frac{2^n - 1}{7}$, then the third term of the sequence is

A. 1
B. 2
C. $\frac{1}{7}$
D. $\frac{2}{7}$

21. The 7th term of the arithmetic sequence 10, 7, 4, 1,…. is

A. – 5
B. – 6
C. – 8
D. – 11

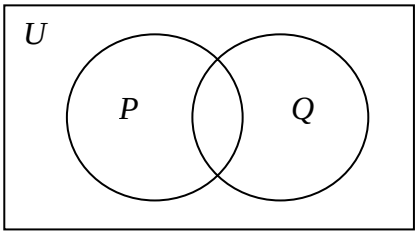
22. The geometric mean between $\frac{1}{7}$ and $\frac{7}{4}$ is

A. $\frac{1}{4}$
B. $\frac{1}{2}$
C. $\frac{7}{2}$
D. $\frac{7}{4}$

23. Which of the following is the set of natural numbers denoted by N ?

A. $\{0,1,2,3,4,.....\}$
B. $\{.....,-2,-1,0,1,2,.....\}$
C. $\{1,2,3,4,5,.....\}$
D. $\{2,3,5,7,11\}$

24. Consider the following Venn diagram. $P \cup (Q \cap P)$ is equal to



A. P
B. Q
C. $P - Q$
D. $Q - P$

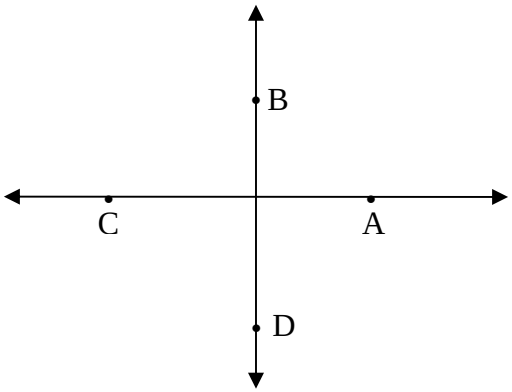
25. $\{\ \} \cup \{1,2,3\} \cap \{3,4\}$ is equal to

A. $\{\ \}$
B. $\{3\}$
C. $\{3,4\}$
D. $\{1,2,3\}$

26. For sets $A = \{1,2,3\}$ and $B = \{a,b,c\}$, $f = \{(a,1),(b,2),(c,3)\}$ is a/an

A. into function from A to B
B. into function from B to A
C. 1 – 1 function from A to B
D. 1 – 1 function from B to A

27. If $a < 0$, then the location of $(a,0)$ in the cartesian plane is



28. Which of the following points satisfies the equation $y = 3x - 2$?

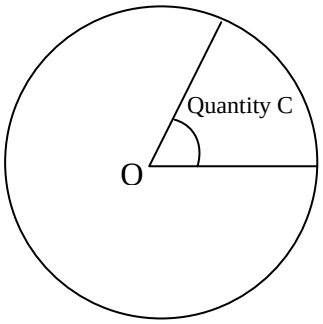
A. $(-5,-1)$
B. $(-1,-5)$
C. $(1,-5)$
D. $(0,1)$

29. The ages, in years, of nine children in a group are 7, 7, 8, 9, 6, 8, 5, 8 and 10. The modal age is

A. 6
B. 7
C. 8
D. 9

30. A quantity *C* is 20% of the whole data. If a pie chart is drawn to represent the data, then the corresponding angle to quantity *C* is

A. 45°
B. 60°
C. 72°
D. 80°



END OF PAPER

Please use this page for rough work